Dermatology Data Set

The data was clean with no Null/NaN values. One of the columns had a trailing space, which I removed. The data was typically scaled using StandardScaler.

Problem 1.1 - Gradient Descent (model1)

The Gradient Descent Algorithm:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{n}\sum_{i=1}^{n}(h(\theta)^i - y^i).\, X_j^i$$

We begin by initializing our model's parameters, represented by the Theta vector (w in the previous notation), with random values. Using these parameters, we predict the output for our input data points using the hypothesis function (often denoted as $h(\theta)^i$).

Next, we quantify how far off our predictions are from the actual target values using a cost function, such as the Mean Squared Error (MSE). The goal of Gradient Descent is to minimize this cost function.

To find the set of parameters that minimizes the cost, we need to know in which direction to adjust our current parameters. This is where the gradient comes in. The gradient is a vector composed of the partial derivatives of the cost function with respect to each parameter (θ_0, θ_1, etc.). The gradient points in the direction of the steepest increase in the cost function. To minimize the cost, we update our parameters by moving in the opposite direction of the gradient. The size of the step we take in that direction is controlled by the learning rate α. The update rule for each parameter is: $\theta_j := \theta_j - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_j}$

The gradient descent update rule for each parameter θ_j is:

$$\theta_j := \theta_j - \alpha \cdot \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta(x^{(i)}) - y^{(i)}\right) x_j^{(i)}$$

Here are the updates for the first few parameters as examples:

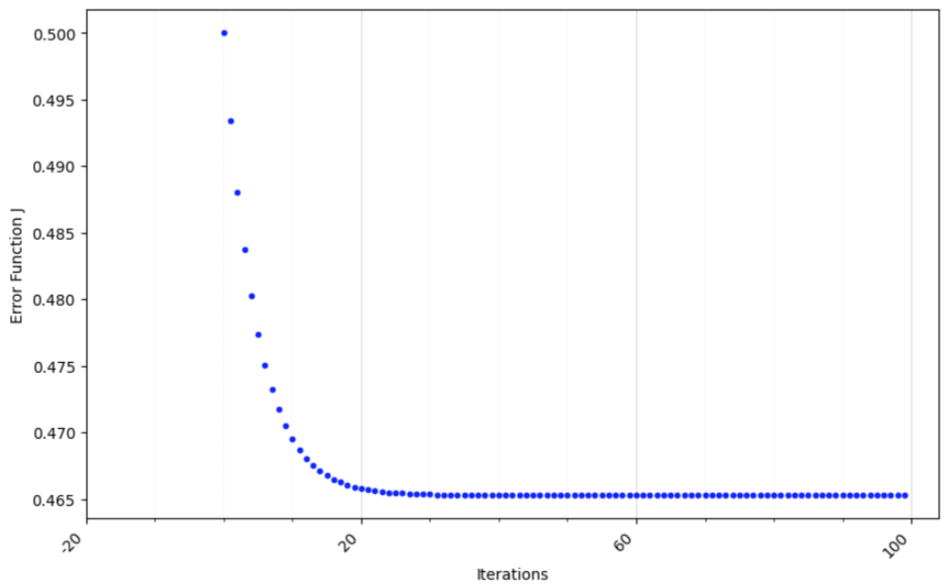
$$\theta_0 := \theta_0 - \alpha \cdot \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta(x^{(i)}) - y^{(i)}\right) x_0^{(i)}$$

$$\theta_1 := \theta_1 - \alpha \cdot \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta(x^{(i)}) - y^{(i)}\right) x_1^{(i)}$$

$$\theta_2 := \theta_2 - \alpha \cdot \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta(x^{(i)}) - y^{(i)}\right) x_2^{(i)}$$

Now that we have the Gradient Descent Algorithm, I first calculated the slope and intercept for Age predicting Disease using Ordinary Least Squares to get slope = 3.7909 and intercept = -.0276. I used these to validate the custom Gradient Descent algorithm.

Plotting Errors by Iterations yields:



As observed from the cost plot, after around 17 iterations the cost is flat. This indicates that the gradient descent algorithm has largely converged by this point, and running for additional iterations would not significantly improve the estimated parameters or reduce the cost further. Therefore, 17 can be considered the approximate optimal number of iterations for convergence in this specific run with the chosen learning rate and initial parameters.

The final estimated parameters for the linear regression model after running gradient descent are:

Intercept: 3.7909

Slope (Age): -0.0276

Ordinary Least Squares (OLS) is another method for finding the best-fitting line in linear regression, and it provides a closed-form analytical solution. When the gradient descent results match the OLS results (or are very close), it indicates that the gradient descent implementation is likely correct and has found the global minimum of the cost function for this linear model.

The resulting linear regression model is:

$$\text{Predicted Disease} = -0.0276 \times \text{Age} + 3.7909$$

The slope of -0.0276 suggests that, according to this model, for every one-year increase in age, the predicted 'Disease' value decreases by approximately 0.0276 units. The intercept of 3.7909 suggests that when the age is zero, the predicted 'Disease' value is approximately 3.7909.

A simple linear regression model probably does not fully capture the complexity of the relationship between age and the type of disease, especially if 'Disease' is not a continuous variable or if the relationship is non-linear. This model provides a basic understanding but may not be accurate for precise disease prediction.

Problem 1.2 - Random Forest (model2)

On the other hand, the Random Forest Model was extremely accurate at 98% (1% point higher after scaling the features; likely due to Age). The ensemble nature of the Random Forest Model with Scaled Features captured complex relationships within the data well, leading to a robust and accurate prediction. The high accuracy suggests high suitability of tree-based methods for this dataset.

Problem 1.3 - KNN (model3)

Similarly, the K Nearest Neighbor (KNN) Model was also very accurate. KNN is a non-parametric, instance-based learning algorithm. At its core, it classifies a new data point based on the majority class among its 'K' nearest data points in the training set. First I tried a range of KNN neighbors from 2 to 9 with accuracy rates ranging from 90% to 94%. I picked k=4 to achieve the best model with 94% accuracy.

The high accuracy (94%) demonstrates that the KNN algorithm is well-suited for this dataset, indicating that data points belonging to the same disease types tend to be close to each other in the feature space. The variation in accuracy across different 'K' values highlights the importance of hyperparameter tuning for KNN; the performance is sensitive to the number of neighbors chosen.

Problem 1.4 - Agglomerative Clustering (model4)

In determining the optimal number of clusters I noted that Elbow point is more defined using unscaled features, which isn't really an indicator of how good or bad the Elbow chart is, because unscaled features might show artificially more distinction in the clustering. The Elbow chart indicated that 4 was the optimal number of clusters.

Next I tried each of the different Agglomerative Clustering methods with 4 clusters and noticed that the ‘ward’ model was the best:

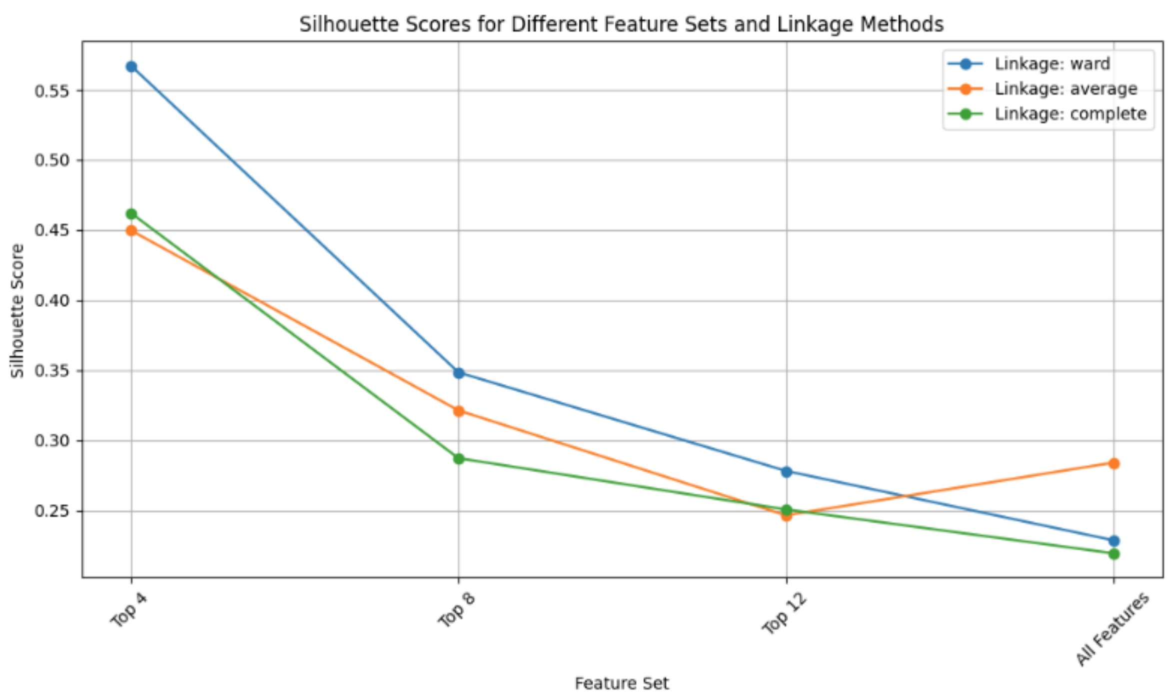
```
Silhouette Score: 0.3014 with linkage_method='ward'
Silhouette Score: 0.3007 with linkage_method='complete'
Silhouette Score: 0.2767 with linkage_method='average'
```

```
Silhouette Score: 0.1749 with linkage_method='single'
```

Finally I wanted to see which features might make the best model, so I used a Decision Tree Classifier to rank the most important features:

```
Clubbing           0.307903
Vacuolisation      0.221932
Fibrosis           0.166393
Koebner            0.117474
Follicular         0.080201
Disapperance       0.033064
Thinning           0.018894
Age                0.014621
Perifollicular     0.013784
PNL                0.009419
Exocytosis         0.006877
Itching            0.005617
Hyperkeratosis     0.003822
Erythema           0.000000
…
```

Unfortunately, limiting to the top 13 features did not improve the model. So then I grouped the features together in 4, 8, 12 and was able to find a much better model:



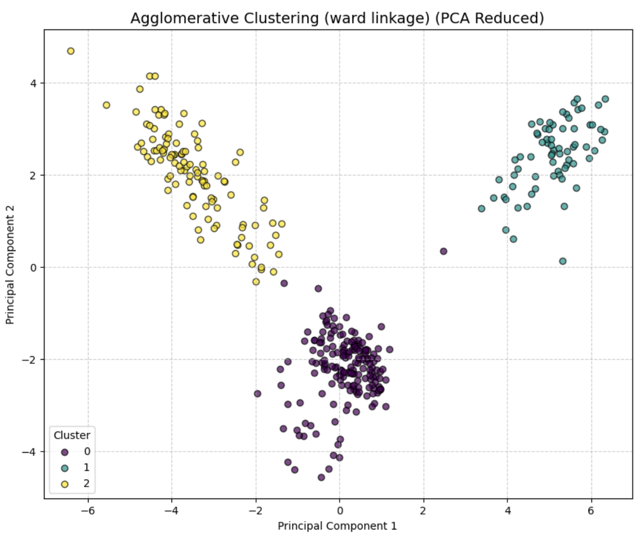
The Silhouette Score for Agglomerative Clustering, using the top 4 features ('Clubbing', 'Vacuolisation', 'Fibrosis', 'Koebner') with linkage_method='ward' was ~.49.

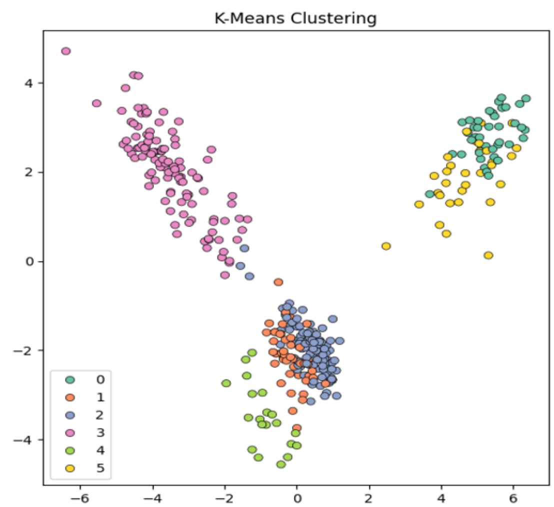
Problem 1.4 - K-Means (model5)

Next I ran a K-Means/Divisive Clustering model with N Clusters of 4 determined by the Elbow Method. This yielded an inertia, or the Sum of Squared Errors, of ~378 and a Silhouette Score of ~.56 which is a moderately positive value. This suggests that, on average, the data points are reasonably well-matched to their assigned clusters and are somewhat separated from neighboring clusters. It indicates that the 4 clusters produced by KMeans have a fair degree of separation and compactness.

Problem 1.4 - Comparison of Agglomerative and K-Means

The visualization of the two clustering methods using Principal Component Analysis (PCA) showed very distinct clustering with Agglomerative, but much more overlap with K-Means.





Problem 1.5 Analysis

Having built and evaluated five different models on the dermatology dataset, a clear picture emerges regarding the data's structure and the suitability of various machine learning techniques for determining disease type.

Model 1, a Simple Linear Regression predicting disease type based solely on Age using Gradient Descent, demonstrated very poor performance with R-squared = 0.069: The model explains only 6.9% of the variance in the dependent variable (Disease). That's extremely low—suggests a poor fit. This outcome was expected, as regression is fundamentally ill-suited for predicting discrete, categorical outcomes like disease types, and Age alone appears to have limited linear correlation with the specific disease categories.

Model 2 and Model 3 perform much better because they are designed for this type of problem. Comparing the classification models, the Random Forest classifier (**Model 2**), trained on all clinical and histopathological attributes, achieved remarkably high accuracy (98%). Its confusion matrix revealed near-perfect classification for several disease types (Psoriasis, Lichen Planus, Chronic Dermatitis, Pityriasis Rubra Pilaris), with only minor confusion primarily between Seboreic Dermatitis and Pityriasis Rosea. The K-Nearest Neighbors classifier (**Model 3**), also using all attributes (scaled), achieved a lower accuracy (93% with k=2). Its confusion matrix showed more significant misclassifications and confusion across multiple classes, indicating that while the features contain predictive information, KNN's simple distance-based approach is less effective than Random Forest's ensemble of decision trees at capturing the potentially complex, non-linear boundaries between these disease types in the feature space.<p>

Model 4: Agglomerative, Model 5: K-Means explored the inherent grouping within the data in an unsupervised manner. The Elbow method on inertia suggested around 3 clusters for both algorithms. While Agglomerative Clustering with the 'average' linkage achieved the highest Silhouette Score among tested linkages (~0.29), this score suggests only a weak clustering structure with some overlap between groups. The dendrogram of disease centroids provided a visual hierarchy of disease similarity based on average feature profiles, but the clustering did not perfectly align with the 6 known disease types. So I used a DecisionTree Classifier to determine feature importance and then plotted different sets of features to see which had the best Silhouette score; it ended up being 'Clubbing', 'Vacuolisation', 'Fibrosis', 'Koebner' so I used those in the target model which yielded a much better score of ~.49.

With ~.46 Silhouette Score for **KMeans** using the top 4 features, clusters exist, they're well-defined, and align well with disease labels. Like with Agglomerative, when using all the features, the Sillouhette score drops by about 50%.

The high performance of the supervised **Random Forest Model2, is the clear winner**. It demonstrates that these features do contain sufficient information to accurately distinguish the known disease types when guided by the actual labels during training.

The analysis conclusively shows that the clinical and histopathological features are highly effective for classifying dermatological diseases. High accuracies achieved by models like Random Forest and KNN confirm their ability to capture complex data relationships. While clustering with KMeans and Agglomerative methods showed some data structure and moderate alignment with known diagnoses, the high performance gain when using multiple features over the age-only Logistic Regression model highlights the value of doing feature selection. Both classification and clustering approaches offer valuable ways to model this data, depending on the specific project goals.

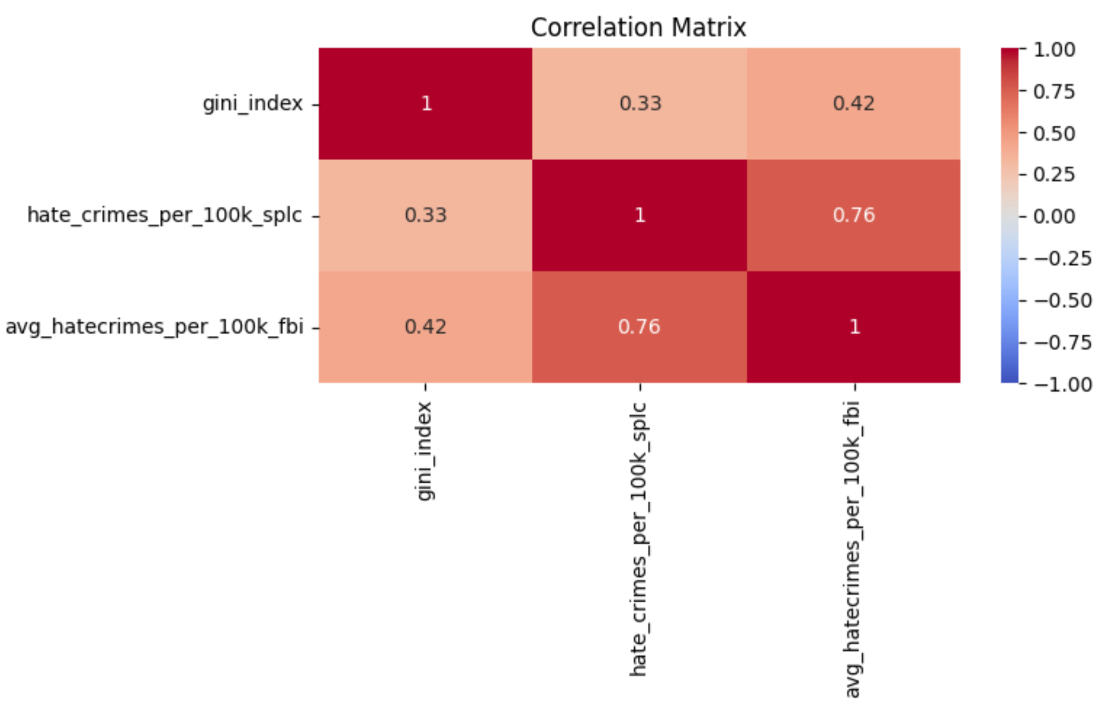
Crime Data Set

The data contained some NaN and null values which were removed (rows). The features were typically scaled with StandardScaler.

Problem 2.1 - Income Inequality and Hate Crimes

"Gini" refers to a the statistician Corrado Gini. In 1912 Gini came up with a number that represents how much inequality of income distribution exists in a given area. This was used to determine the correlation between the SPLC and FBI data.

Gini vs SPLC: 0.329 - Income inequality is moderately associated with reported hate crimes (SPLC data). Gini vs FBI: 0.421 - The association is even stronger with official FBI reports. As you can see from the heatmap, there is a moderate positive correlation between income inequality and hate crime rates. The Gini index correlates with SPLC-reported hate crimes at 0.329 and with FBI-reported rates at 0.421, suggesting that states with greater income inequality tend to report more hate crimes per 100,000 people.



Problem 2.2 - Population and Hate Crimes

I ran a basic Ordinary Least Squares Model on the features given in the data set first predicting the SPLC hate crime rate and then the FBI hate crime rate. The OLS model predicting SPLC explains more variance with a higher R^2 (.56 vs .60) and Adj. R^2 (.44 vs .49). It has a better AIC and log-likelihood, showing a better fit despite the small sample size. `gini_index` and `share_voters_voted_trump` are significant in this model. The model predicting FBI only gini_index is clearly significant and it's R^2 scores are worse than the SPLC model.

Measurement	Predict SPLC	Predict FBI
R^2	.60	.56
Adjusted R^2	.49	.44
AIC	-18	159

Since several features are not siginficant (P<.05), I removed them and ran the SPLC model again with the following results:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     hate_crimes_per_100k_splc   R-squared:                       0.441
Model:                                   OLS   Adj. R-squared:                  0.414
Method:                        Least Squares   F-statistic:                     16.55
Date:                       Wed, 07 May 2025   Prob (F-statistic):           5.02e-06
Time:                               21:09:15   Log-Likelihood:                 11.830
No. Observations:                         45   AIC:                            -17.66
Df Residuals:                             42   BIC:                            -12.24
Df Model:                                  2
Covariance Type:                   nonrobust
============================================================================================
                               coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------
const                        0.2788      0.770      0.362      0.719      -1.275       1.833
gini_index                   1.4582      1.539      0.948      0.349      -1.647       4.563
share_voters_voted_trump    -1.3290      0.282     -4.714      0.000      -1.898      -0.760
==============================================================================
Omnibus:                        6.565   Durbin-Watson:                   1.737
Prob(Omnibus):                  0.038   Jarque-Bera (JB):                5.469
Skew:                           0.812   Prob(JB):                       0.0649
Kurtosis:                       3.526   Cond. No.                        72.0
==============================================================================
```

To predict the continuous value of hate crimes per capita, several regression models were tested and evaluated using the R-squared metric.

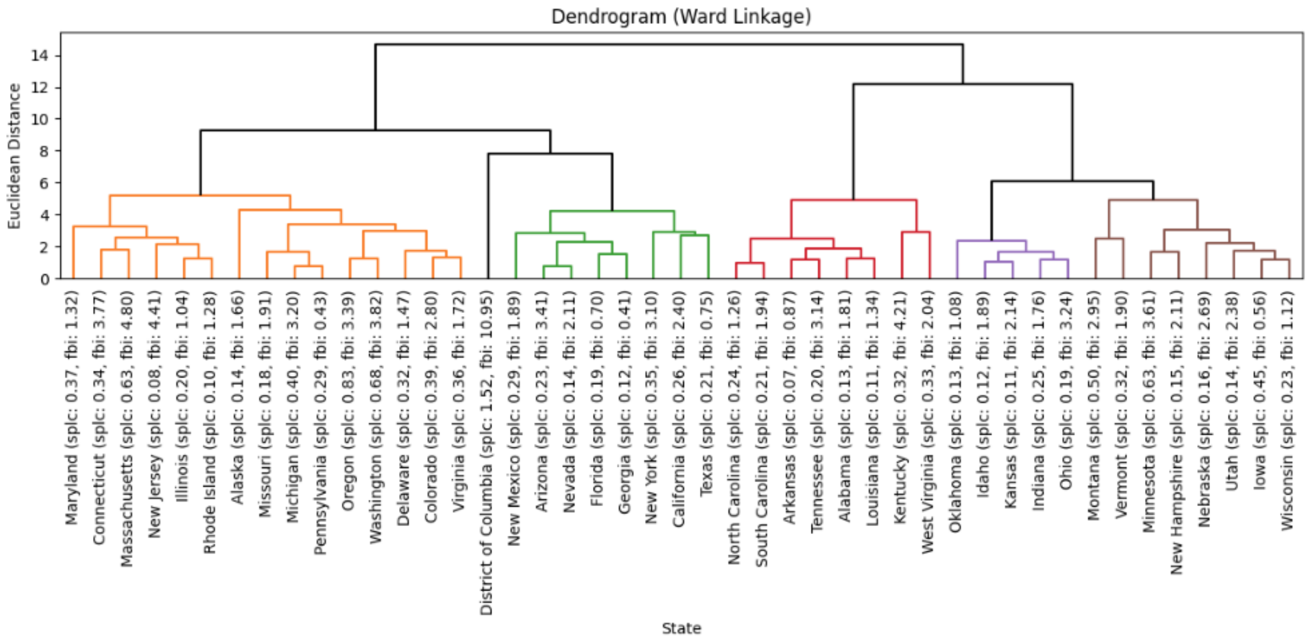
Initial attempts with a K-Nearest Neighbors (KNN) Regressor and a Random Forest Regressor yielded low R-squared values (approximately 0.09 and 0.12, respectively, for the better-performing SPLC data), indicating poor predictive power. A Random Forest model predicting the FBI data even resulted in a negative R-squared (~-0.10), performing worse than simply predicting the mean.<p>

Ordinary Least Squares (OLS) regression showed significantly better results. Predicting avg_hatecrimes_per_100k_fbi with OLS gave an R-squared of approximately 0.56. The performance improved when predicting hate_crimes_per_100k_splc with OLS, achieving an R-squared of approximately 0.60. An attempt to simplify the OLS model by removing non-statistically significant features (based on P-values) resulted in a reduced R-squared of approximately 0.44, demonstrating that including all features provided a better fit in this case.<p>

Therefore, the Ordinary Least Squares (OLS) model, utilizing all features to predict hate_crimes_per_100k_splc, was identified as the best-performing model among those evaluated, with an R-squared of approximately 0.60.

Problem 2.3 - Hate Crimes Across States

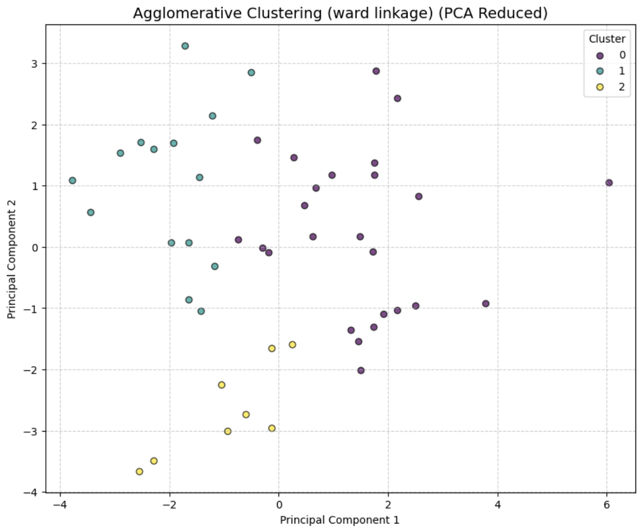
Preparing a Dendogram with ward linkage, provided some initial insights into the state by state comparison: we immediately notice that Washington DC is an outlier. Being the smallest “state” in the US, this is not surprising. I considered removing it from the data set as an outlier, but decided to keep it instead.

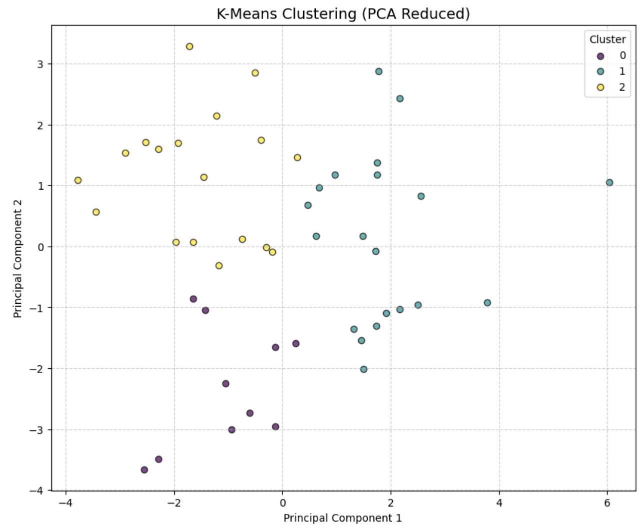


The following scatter plots visualize the results of applying Agglomerative Clustering with Ward linkage and KMeans Clustering, both with PCA, to the states. The plot shows how the states are nicely grouped into 3 clusters (labeled 0, 1, and 2). Visually, the clusters appear mostly separated, suggesting that based on the features used, states can be reasonably divided into three distinct groups.

The Agglomerative Cluster 0 seems to contain a larger, more spread-out group, while Clusters 1 and 2 represent more concentrated groupings in different areas of the PCA space.

The KMeans plot shows the states grouped into 4 clusters (labeled 0, 1, 2, and 3). Similar to the Agglomerative clustering, the clusters show some visual separation, indicating different groupings of states based on their features.





Cluster 0: ['AL', 'AR', 'IN', 'KY', 'LA', 'NC', 'OK', 'SC', 'TN', 'WV']
Cluster 1: ['AK', 'CO', 'ID', 'IA', 'KS', 'MI', 'MN', 'MO', 'MT', 'NE', 'NH', 'OH', 'PA', 'UT', 'VT', 'WI']
Cluster 2: ['AZ', 'CA', 'CT', 'DE', 'DC', 'FL', 'GA', 'IL', 'MD', 'MA', 'NV', 'NJ', 'NM', 'NY', 'OR', 'RI', 'TX', 'VA', 'WA']

The following two tables represent the average predicted values for each Cluster type (Aggolomerative on the left and K-Means on the right)

Cluster	hate_crimes_per_100k_splc	avg_hatecrimes_per_100k_fbi
0	0.359240	2.615797
1	0.259696	2.111143
2	0.201443	2.074693

Cluster	hate_crimes_per_100k_splc	avg_hatecrimes_per_100k_fbi
0	0.199218	1.943657
1	0.275192	2.162972
2	0.379689	2.777759